

ASSIGNMENT - DAY 3

RAM and Storage

Theory: Types of RAM (DDR3, DDR4, DDR5) and storage (HDD vs. SSD, NVMe).

Practical: Remove and reinstall RAM and storage drives.

Task - 1

Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
Typical Speed	800–2133 MT/s	1600–3200 MT/s	4800 MT/s and above
Voltage	About 1.5 V	About 1.2 V	About 1.1 V
Typical Use	Older desktops and laptops	Modern desktops and laptops	Newer/high-performance PCs and laptops

Conclusion: DDR3 is an older generation, DDR4 improves speed and power efficiency, and DDR5 is a newer generation designed for higher memory performance.

Task - 2

Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

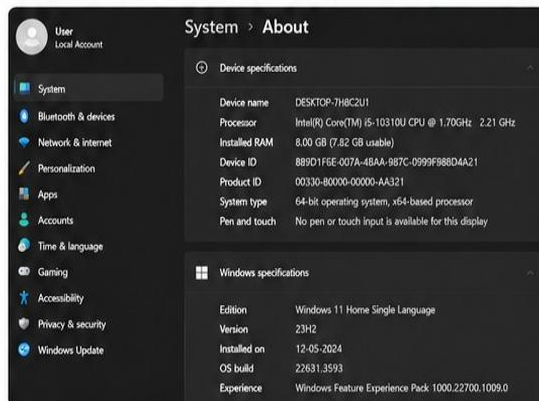
Reference Image – RAM Module and Boot Verification

Photo of Installed RAM Module:



Reinstalled RAM and System Boot Verification:

After reinstalling the RAM module, I turned on the system and it booted up successfully.



Reference visual only; use your own practical photo if required.

Practical Steps: Shut down and disconnect power, locate the RAM slot, remove the module carefully, reinstall it in the correct orientation until the clips lock, then start the system and verify that the RAM is detected and the computer boots.

Task - 3

List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	HDD	SSD	NVMe
Speed	Slowest	Faster than HDD	Generally fastest
Durability	Moving parts; more sensitive to shock	No moving parts; more shock-resistant	No moving parts; high performance
Typical Use	Large storage and backups	OS, apps and gaming	Gaming and heavy workloads
Example	Photo/video backups	Windows and daily apps	Large games / video editing

HDD: Best suited to large storage and backups where speed is less important.

SSD: Good for the operating system and everyday applications because it is faster than an HDD.

NVMe: Good for high-performance tasks such as gaming, video editing, and applications that benefit from fast loading.

Task - 4

Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Reference Photo – Storage Drive, Connector and Recognition



The reference image shows a SATA SSD connector and an example of a drive being recognized by the system.

Practical Steps: Shut down the system and disconnect power. Identify the storage drive and connector, remove the drive carefully, reinstall it securely, then start the system and confirm that the storage is recognized.

Connector Type: SATA / M.2 / NVMe — select the correct type after checking your own device.

Important: Do not claim the reference photo is your own practical evidence. If your teacher requires your actual photo, replace it with a photo of your device.